Geometry
Unit 2 Assessment

Name____________________
Date__________________
Period__________

Determine the type of reasoning for each statement.

	Statement	Type of Reasoning
1.	Sofia notices that the dogs on her street bark at a delivery truck, so Sofia concludes that all dogs bark at delivery trucks.	Inductive
2.	Florida requires all lawyers to pass the bar exam to practice. If Mark does not pass the bar, then he will not legally be able to represent as their lawyer.	Deductive
3.	The sum of the measures of two angles is $90°$, then the angles are complementary.	Deductive

4. An incomplete proof is shown. Complete the reason for step 2.

Statement	Reason
1. $m\angle TWX = m\angle YRG$	1. Given
2. $\angle TWX \cong \angle YRG$	2. Definition of Congruence

5. A series of steps for playing UNO is shown. The steps are not possible.

Steps
1. Green 7
2. Red 7
3. Yellow 9
4. Yellow 3
5. Wild Card
6. Red Reverse

a. Which step contains the error?

Error	Step **3**

b. What card, other than a wild card, would correctly complete the steps?

Card	Yellow 7 or Red 3

6. A statement is shown.

"If it is my birthday, then I will eat ice cream."

a. Write the converse of the statement.

Converse	If I eat ice cream, then it is my birthday.

b. Write a counterexample that shows the converse is not true.

Counterexample	*Answers will vary.* I eat ice cream on my brother's birthday.

7. Write each statement as an if-then statement.

Statement	If-Then
All Floridians have been to the beach.	If someone is a Floridian, then they have been to the beach.
All four angles at the point of intersection of two lines are right angles.	If two lines intersect, then all four angles at the point of intersection are right angles.

8. Complete the table for the conditional statement shown.

If Anthony has visited the Southernmost Point Buoy, then he has been to Key West.

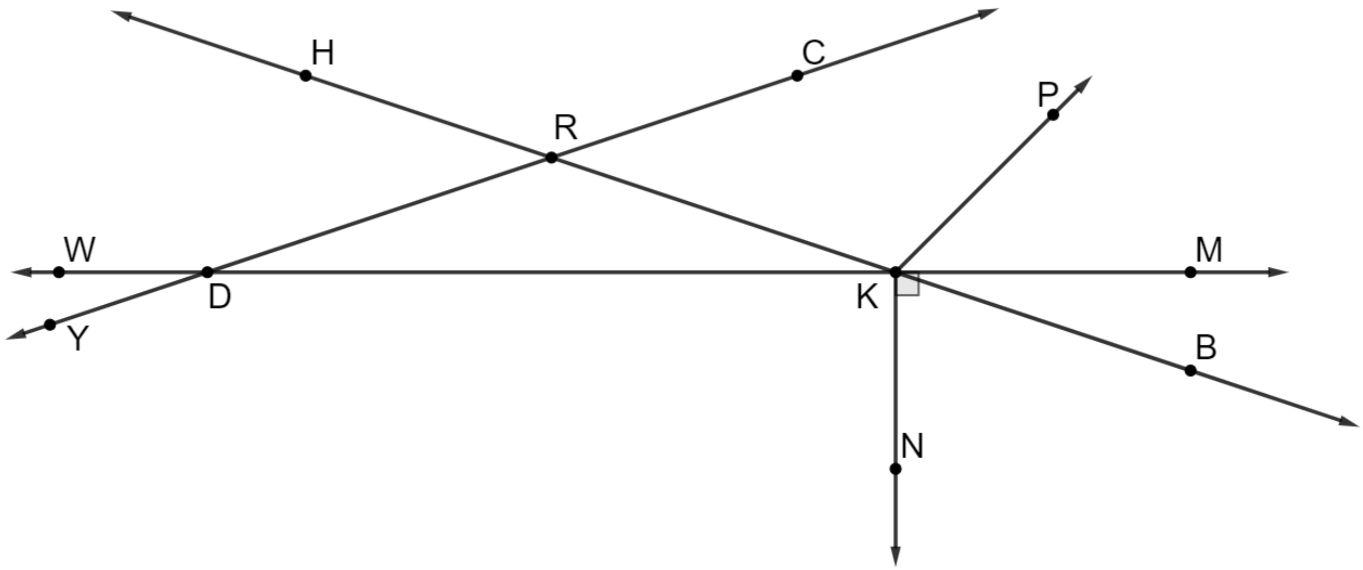
Hypothesis	Anthony has visited the Southernmost Point Buoy.
Conclusion	Anthony has been to Key West.

9. Write each conditional using "if and only if."

If the sum of two angles is $180°$, then the angles are supplementary.

Biconditional	Two angles are supplementary if and only if their sum is $180°$.

10. A diagram is shown.



a. Select all of the statements that are true.

- ☐ $\angle DRH \cong \angle KRC$
- ☐ $\angle HRC \cong \angle KRD$
- ☐ $\angle MKP \cong \angle BKN$
- ☐ $\angle NKB \cong \angle RKP$
- ☐ $\angle RDK \cong \angle RKD$
- ☐ $\angle WDY \cong \angle RDK$

b. An incomplete proof is shown. Complete the reasons for steps 2, 3, and 4.

Given: $m\angle MKB + m\angle BKN = 90°$

Prove: $m\angle RKD + m\angle BKN = 90°$

Statement	Reason
1. $m\angle MKB + m\angle BKN = 90°$	1. Given
2. $\angle MKB \cong \angle RKD$	2. Vertical Angles Theorem
3. $m\angle MKB = m\angle RKD$	3. Definition of congruence
4. $m\angle RKD + m\angle BKN = 90°$	4. Substitution

c. Complete the table for the conditional statement.

If $m\angle MKN = 90°$, then $m\angle MKB + m\angle BKN = 90°$.

Relationship to Conditional	Statement
Converse	If $m\angle MKB + m\angle BKN = 90°$, then $m\angle MKN = 90°$.
Inverse	If $m\angle MKN \neq 90°$, then $m\angle MKB + m\angle BKN \neq 90°$.
Contrapositive	If $m\angle MKB + m\angle BKN \neq 90°$ then $m\angle MKN \neq 90°$.

d. Explain why the conditional statement is biconditional.
Both the statement and the converse are true.

11. A series of steps is possible in the game of UNO. Each statement represents a card played.

Statement	Reason
1. Blue 5	1. Given
2. Blue 2	2. Matches color
3. Yellow 2	3. Matches number

Without using a wild card, add two additional statements and provide the reason for the series of steps.

Statement	Reason
4. *Answers will vary*	4. *Answers will vary*
5. *Answers will vary*	5. *Answers will vary*